

The Effective String of 3D Gauge Systems at the Deconfining Transition

Source-backed arXiv sample

1 Extracted Lists

The list environments below are extracted from arXiv LaTeX source and wrapped for pdfLaTeX validation.

SU(2)] In this case there is a good convergence among different published results both for the string tension and for the critical temperature. Data on T_c are reported in [?] for lattices with length $L = 2$ in the imaginary time direction; in [?] for $L = 4$ and in [?] for $L = 5$ and 6. Scaling seems to be fulfilled already for $L > 4$ and allows a rather precise estimation: $T_c \beta a = 1.4 \pm 0.1$ [?] (where a is the lattice spacing and β the inverse of the coupling constant). Data on σ are reported in [?] and more recently in [?], in both case obtained with Wilson loops. Scaling seems to be fulfilled already for $\beta > 6$. (in agreement with the T_c behaviour). These two sets of data agree and give $\sigma(\beta a)^2 = 2.23 \pm 0.04$. This implies $T_c/\sqrt{\sigma} = 0.94 \pm 0.03$ in agreement with our critical temperature for the minimal radius r_1 , $T_c/\sqrt{\sigma} = \sqrt{3}/\sqrt{\pi} = 0.98$. Indeed this result was already presented in [?] to support the Nambu-Goto string picture, and has not been significantly changed by the new high precision data of ref [?].

Z₂] In this case a rather precise estimate exists for the string tension [?] : $\sigma a^2(\beta_c - \beta)^{-2\nu} = 3.64 \pm 0.09$, with $\beta_c = 0.7614$ and $\nu = 0.63$. For the critical temperature, the scaling region seems to start from $L = 4$. Data obtained with the Montecarlo renormalization group approach [?] for $L = 4, 8$, give $T_c a(\beta_c - \beta)^{-\nu} = 2.3 \pm 0.1$, while a slightly lower value was obtained using finite size scaling techniques [?, ?] for $L = 4$: $T_c a(\beta_c - \beta)^{-\nu} = 2.2 \pm 0.1$. Taking into account all the uncertainties we can assume: $T_c/\sqrt{\sigma} = 1.17 \pm 0.1$ which is in acceptable agreement with our solution.